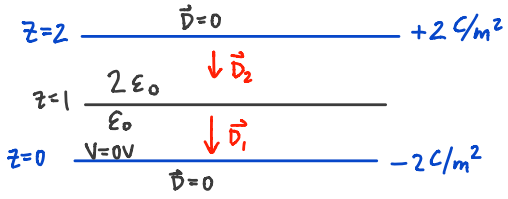
1. Two parts of this question are independent:

a) (10 pts) Consider two infinite conducting plates with facing surfaces on $z = 0$ and $z = 2$ m planes. The surfaces contain ± 2 C/m^2 of charge density, the top plate being positively charged. In between the plates the region $z > 1$ is filled with an ideal dielectric with permittivity $\epsilon = 2\epsilon_o$ while $z < 0$ is free space. On $z = 1$ m surface of the dielectric there is no surface charge.

[z<1] ?

i. (5 pts) If the potential of the bottom plate is assigned 0 V, that is $V(0) = 0$ V, determine the potentials $V(1)$ and $V(2)$ at $z = 1$ m and $z = 2$ m, respectively. Show your work.



$\hat{n}\cdot(\vec{D}_1 - \vec{D}) = -2$ C/m^2

$\vec{D}_1 = -2\hat{z}$ C/m^2 $\rightarrow$ $\vec{E}_1 = \frac{-2}{\epsilon_0}\hat{z}\ \frac{V}{m}\cdot 1\text{m} \rightarrow \Delta V = \frac{2}{\epsilon_0}V$

$\vec{D}_2 = \vec{D}_1 = -2\hat{z}$ C/m^2 $\rightarrow$ $\vec{E}_2 = -\frac{1}{\epsilon_0}\hat{z}\ \frac{V}{m}\cdot 1\text{m} \rightarrow \Delta V = \frac{1}{\epsilon_0}V$

$V(1) = 0 + \frac{2}{\epsilon_0}V$, $V(2) = V(1) + \frac{1}{\epsilon_0}V$

$V(1) = \frac{2}{\epsilon_0}V$
$V(2) = \frac{3}{\epsilon_0}V$

ii. (5 pts) What are the explicit values of **E**, **D**, and **P** in $z < 1$ m and $z > 1$m regions in between the plates?

Region 1 ($0 < z < 1$)
$\vec{D}_1 = -2z$ C/m^2
$\vec{E}_1 = -\frac{2}{\epsilon_0}\hat{z}\ \frac{V}{m}$
$\vec{P}_1 = 0$ C/m^2 (free space)

Region 2 ($1 < z < 2$)
$\vec{D}_2 = -2\hat{z}$ C/m^2
$\vec{E}_2 = -\frac{1}{\epsilon_0}\hat{z}\ \frac{V}{m}$
$\vec{P}_2 = -1\hat{z}$ C/m^2

$\vec{D} = \epsilon_0\vec{E} + \vec{P}$
$\vec{P} = \vec{D} - \epsilon_0\vec{E}$

iii. (5 pts) What is the capacitance per unit area of the configuration described above? **Hint:** capacitance is charge per voltage difference.

$C = \frac{Q}{\Delta V} = \frac{2\text{ C/m}^2}{3/\epsilon_0 V} = \frac{2\epsilon_0}{3}$

$C = \frac{2\epsilon_0}{3}\ \frac{F}{m^2}$

b) In a homogeneous conductor where $\mathbf{J} = \sigma\mathbf{E}$, Gauss's law $\nabla\cdot\mathbf{E} = \frac{\rho}{\epsilon_o}$ and the continuity equation $\frac{\partial\rho}{\partial t} + \nabla\cdot\mathbf{J} = 0$ can be used together to derive a differential equation

$$\frac{\partial\rho}{\partial t} + \frac{\sigma}{\epsilon_o}\rho = 0$$

for the charge density ρ.

i. (5 pts) Find the solution of the differential equation above for $t > 0$ if at $t = 0$ the charge density is $\rho(x, y, z, 0) = \sin(4z)$ C/m^3 over all space.

$\rho(0) = \sin(4z)$ C/m^3

$\frac{\partial\rho}{\partial t} = -\frac{\sigma}{\epsilon_0}\rho \rightarrow \rho(t) = \rho(0)\,e^{-\frac{\sigma}{\epsilon_0}t}\ \frac{C}{m^3}$

$\rho(x,y,z,t) = \sin(4z)\,e^{-\frac{\sigma}{\epsilon_0}t}$ C/m^3

ii. (5 pts) For $\rho(x, y, z, t)$ given in part (i) what would be the corresponding electric field **E** at $t = 0$ and as $t \rightarrow \infty$ if conductivity $\sigma > 0$? **Hint:** First determine $\partial E_z/\partial z$ and then find its anti-derivative.

$\nabla\cdot\vec{E} = \frac{\rho}{\epsilon_0}$

$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \frac{\sin(4z)\,e^{-\frac{\sigma}{\epsilon_0}t}}{\epsilon_0}$

$E_z = \frac{-\cos(4z)\,e^{-\frac{\sigma}{\epsilon_0}t}}{4\epsilon_0}\ \frac{V}{m}$

At $t=0$:
$\vec{E}(x,y,z,0) = \frac{-\cos(4z)}{4\epsilon_0}\hat{z}\ \frac{V}{m}$

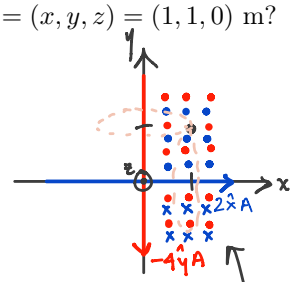
As $t\rightarrow\infty$
$\vec{E}(x,y,z,t) = 0\ \frac{V}{m}$

2. Two parts of the this question are independent:

a) Consider the current density

$$\mathbf{J}(x,y,z) = \overbrace{2\hat{x}\,\delta(y)\delta(z)}^{\text{line charge of } 2\hat{x}\text{ A}} - \underbrace{4\hat{y}\delta(x)\delta(z)}_{\text{line charge of } -4\hat{y}\text{ A}} \frac{\text{A}}{\text{m}^2}.$$

i. (5 pts) What is the DC magnetic flux density $\mathbf{B}(1,1,0)$ that is produced by $\mathbf{J}$ at position $\mathbf{r} = (x,y,z) = (1,1,0)$ m?



$B_z \cdot 2\pi = \mu_0(4)$

$B_z = \frac{2\mu_0}{\pi}$ T

$B_z \cdot 2\pi = \mu_0(2)$

$B_z = \frac{\mu_0}{\pi}$ T

SUPERPOSITION: $\left(\frac{2\mu_0}{\pi} + \frac{\mu_0}{\pi}\right)\hat{z}$ T

$\vec{B}(1,1,0) = \frac{3\mu_0}{\pi}\hat{z}$ T

ii. (5 pts) Find a location (x,y,z) where $\mathbf{B}$ produced by $\mathbf{J}$ is identically zero. Explain carefully.

They can cancel out in this region, since the fields point in opposite directions.

$B_z = \frac{\mu_0 4}{2\pi r} = \frac{2\mu_0}{\pi r_1} = B_z = \frac{2\mu_0}{2\pi r} = \frac{\mu_0}{\pi r_2}$

r_1 needs to be twice r_2 → x needs to be 2y (in quad. II)

$(-2,-1)$

At $(-2,1,0)$, $\vec{B} = 0$ T

b) Consider a long solenoid around and parallel to x-axis having $N = 1000$ loops/meter and a radius of $a_1 = 1$ m.

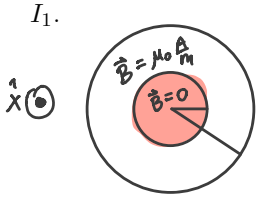
i. (5 pts) If the solenoid current is $I_1 = 1$ mA and this generates an internal magnetic field intensity $\mathbf{H} = H_o\hat{x}$ determine H_o (with the appropriate unit)

N = 1000 Inf: ? $\vec{H} = IN\hat{x}$ A/m² $= (1\times10^{-3}\text{ A})(10^3 \text{ loops/m})\hat{x} = 1\hat{x}$ A/m



$H_0 = 1\ \frac{\text{A}}{\text{m}}$

ii. (5 pts) Another solenoid with $N = 500$ loops/meter and a radius $a_2 = 0.5$ m is inserted within the first solenoid and then its current I_2 is adjusted so that $\mathbf{H} = 0$ for $r = \sqrt{y^2+z^2} < a_2$. Describe I_2 needed to accomplish this in comparison to the magnitude and direction of I_1.



$H_2 = (500)(I_2) = -H_1 = -1\hat{x}$ A/m

$I_2 = -\frac{1}{500} = -\frac{2}{1000}$ A $= -2$mA

$I_2 = -2$mA

iii. (5 pts) If I_1 slowly decreases in time so that H_o decreases in proportion, how will I_2 respond to this change according to Faraday's law and Lenz's rule - briefly describe.

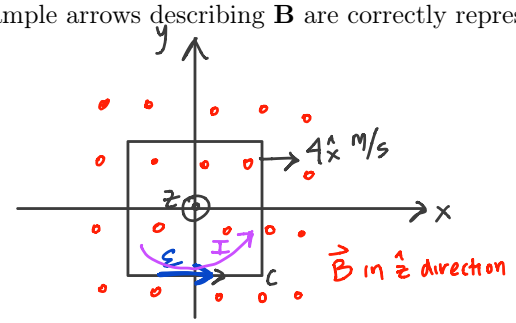
There will be an increase in flux in the $-\hat{x}$ when $\vec{H}_2$ begins to overpower the decaying $\vec{H}_1$. Due to this, there will be an induced current in I_2 such that a field is generated in the $+\hat{x}$ direction to compensate. This induced current opposes the original I_2, and therefore I_2 will decrease with time as well (it will become more positive I should say... it will not decrease to something more negative than -2mA, it will decrease towards zero and possibly become positive depending on the behavior of I_1.

3. A conducting square wire loop of dimensions 1 m × 1 m is moved with velocity $\mathbf{v} = 4\hat{x}$ m/s in a region where the background magnetostatic field is described by

$$\mathbf{B}(x, y, z) = \hat{z}\, 25 \times 10^{-6}(1 + x/L)\,\mathrm{T},$$

where $L = 1000$ m. The center of the loop coincides with the origin $(x, y, z) = (0, 0, 0)$ at $t = 0$ and the loop is oriented so that its flux linkage is maximized for the given static $\mathbf{B}$ field.

a) (5 pts) Draw a diagram where the orientation and motion direction of the square loop and some sample arrows describing $\mathbf{B}$ are correctly represented.

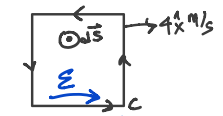


b) (5 pts) Determine the flux linkage $\Psi(0)$ of the square loop at time $t = 0$, assuming $d\mathbf{S} = dx\, dy\, \hat{z}$ in flux calculation.

$$\Psi = \int_S \vec{B} \cdot (dx\,dy\,\hat{z}) = \vec{B}(1\,m^2) = B = 25\times10^{-6}\left(1+\frac{x}{L}\right)\Big|_{x=0}\ [Wb]$$

$$\boxed{\Psi(0) = 25\times10^{-6}\ Wb}$$

c) (5 pts) Determine the emf $\mathcal{E}(0)$ at time $t = 0$ and indicate the emf direction with an arrow shown in the diagram in part (a). **Hint:** be careful! this is a very easy problem but it is also easy to get it wrong!



This is the direction of a positive EMF, assigned to agree with C and $d\vec{S}$

$$\mathcal{E} = -\frac{\partial\Psi}{\partial t} = -\left[\frac{25\times10^{-6}}{L}\cdot\frac{dx}{dt}\right]_{4\hat{x}\ m/s} = -\frac{25\times10^{-6}}{10^3}\cdot 4 = -100\times10^{-9}\ Wb$$

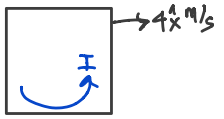
$$\boxed{\mathcal{E}(0) = -10^{-7}\ Wb}$$

d) (5 pts) Is the emf of the loop time dependent? Explain.

No, since the time rate of change of magnetic flux is dependent on the time rate of change of position, and that velocity is constant, then there will be a constant EMF induced by the motion of the loop through this field.

To put it another way, the magnetic flux is changing linearly, so its $\frac{\partial}{\partial t}$ = const.

e) (5 pts) Determine the loop current if the loop resistance is $2\,\Omega$ and show the current and its direction properly in the diagram in part (a)?



$$I = \frac{\mathcal{E}}{R} = \frac{-10^{-7}\ V}{2\,\Omega} = -5\times10^{-7}\ A$$

$$\boxed{I = -5\times10^{-7}\ A}$$

↳ This is the direction for a positive current, since our result was negative, it would actually run clockwise so as to agree with Lenz' Law.